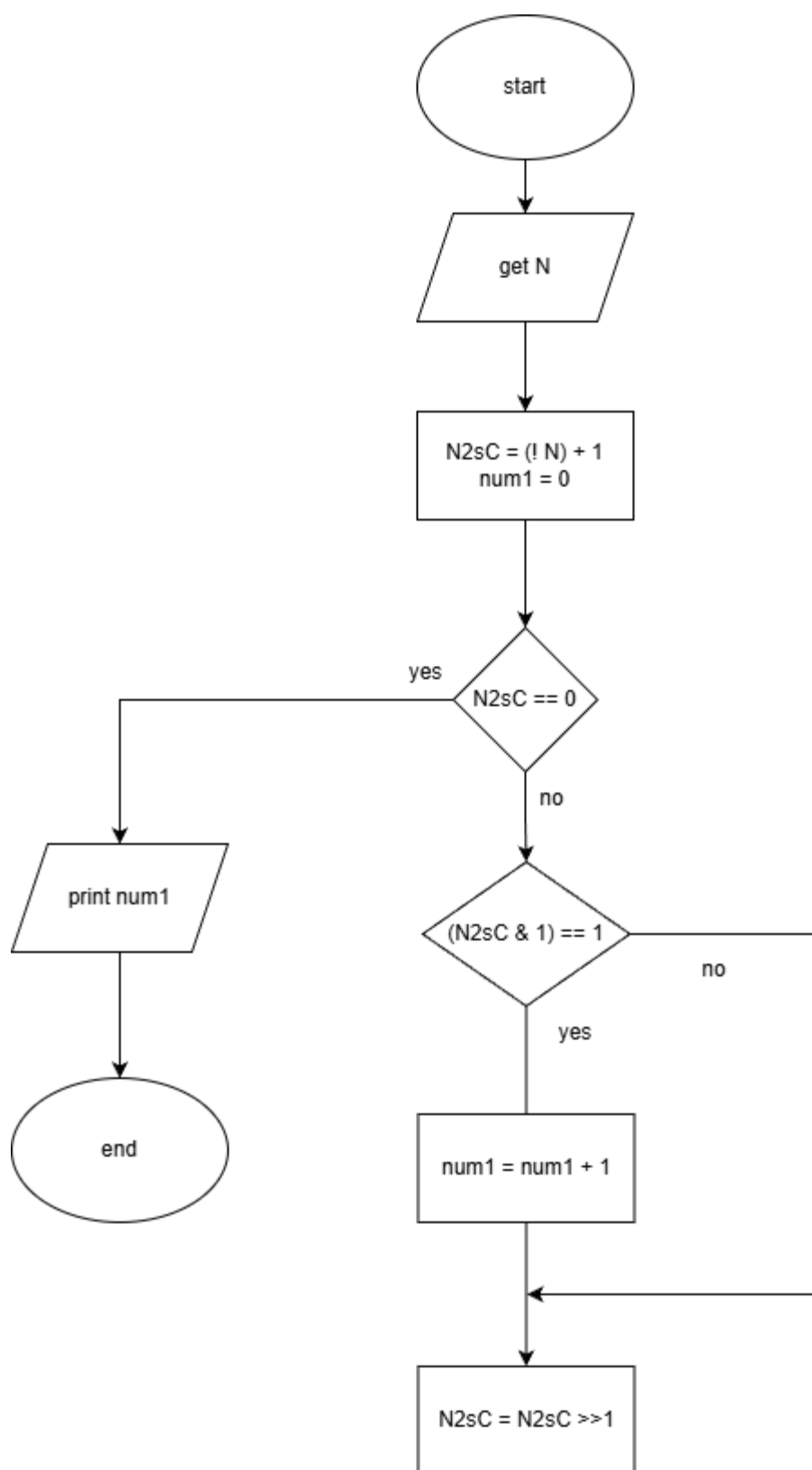




سوال راز الگوریتم خوارزمی

$$N = 10100111 \Rightarrow N_{2SC} = 01011000 + 1 \quad \text{مرحله 2}$$

$$\Rightarrow N_{2SC} = 01011001 \Rightarrow \text{num1} = 1 \Rightarrow N_{2SC} = 00101100$$

$$\Rightarrow N_{2SC} = 00010110 \Rightarrow N_{2SC} = 00001011 \Rightarrow \text{num1} = 2$$

$$\Rightarrow N_{2SC} = 00000101 \Rightarrow \text{num1} = 3 \Rightarrow N_{2SC} = 00000010$$

$$\Rightarrow N_{2SC} = 00000001 \Rightarrow \text{num1} = 4 \Rightarrow N_{2SC} = 00000000$$

$$\Rightarrow \boxed{\text{num1} = 4}$$

$$(421)_5 = 4 \times 5^2 + 2 \times 5 + 1 = 111 \quad \text{مرحله 3}$$

$$111 = 64 + 35 = 64 + 32 + 3 = 64 + 32 + 2 + 1 = 2^6 + 2^5 + 2^1 + 2^0$$

$$(421)_5 = 111 = (1100011)_2 \Rightarrow \boxed{(0000110)_2} = (1100011)_2 \gg 4$$

$$(0000110)_2 = 6 \Rightarrow \boxed{ZIBA} \quad \text{مرحله 4}$$

$$3; \# / \Rightarrow (33, 3B, 23, 2F)_{16} \quad \text{مرحله 5}$$

$$ZIBA \Rightarrow (5A, 49, 42, 41)_{16}$$

Subject: _____

Date _____

404110453

شماره پاسپورت

کد دانشجویی

XOR

0011 0011, 0011 1011, 0010 0011, 0010 1111

0101 1010, 0100 1001, 0100 0010, 0100 0001

0110 1001, 0111 0010, 0110 0001, 0110 1110

$\Rightarrow (69, 72, 61, 6E)_{16} \Rightarrow \boxed{\text{iran}}$